

Deterministic Choice Probabilities and Share Inversion for Factor Probit with Many Alternatives

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August 2026 — working draft, revised

Abstract

Multinomial probit choice probabilities are high-dimensional Gaussian integrals with no closed form; the standard tool, the GHK simulator, computes them one alternative at a time by smoothed importance sampling. For the factor-structured case $\Sigma = VV^\top + \text{diag}(D)$ we describe a deterministic method that computes *all* N probabilities together: conditionally on k Gaussian factors the alternatives are independent, and a lattice survival-product identity yields the entire share vector in $O(QNL)$ for Q quadrature nodes and L lattice points. In the tested $k = 2$ problems the method kept maximum absolute share error below 10^{-3} from $N = 5$ to $N = 5000$ (22 seconds at $N = 5000$, where the Monte Carlo reference’s own noise is of comparable order), and became faster than our reference GHK implementation near $N = 200$; extrapolating that implementation’s timing and $R^{-1/2}$ accuracy suggests hours, not seconds, at matched error for $N = 5000$. Because the approximation is reproducible and differentiable without resampling, it supports practical market-share inversion (a Jacobi-style diagonal-Newton fixed point; $N = 1000$ in under a minute) and reuse of the conditional field for all N single-removal counterfactuals in $O(QN^2L)$. Boundary experiments are reported as they fell: on dense covariances with three spectral-decay profiles, the rank-8 approximation matched or beat GHK at $R = 10^4$ throughout the tested size, and in a substitution study with all noise families standardized to unit variance, factor structure supplied most of the deletion-counterfactual accuracy with the idiosyncratic family contributing a further improvement in this Gaussian-adjacent design.

1 Introduction

The multinomial probit (MNP) model computes choice among N alternatives with jointly normal utilities, allowing flexible substitution patterns — the property that logit-family models trade away for tractability. Its probabilities are $(N - 1)$ -dimensional Gaussian orthant integrals. The field’s workhorse for three decades has been the GHK simulator [1, 2, 3]: smoothed importance sampling along a Cholesky factor, one alternative at a time. GHK is accurate and well understood at moderate N ; its structural costs are that error decays as $R^{-1/2}$ in the number of draws, that the full share vector requires a separate computation per alternative, and that

*Numbers in this paper are produced by committed, seeded scripts with correctness anchors run before any comparison (<https://github.com/microprediction/kinetics>, experiments 13–15; an immutable commit will be pinned at submission). The algorithm ships in the `thurstone` package; package–benchmark agreement is itself a reported number. Benchmarks ran single-threaded NumPy/SciPy (OpenBLAS) on an Apple arm64 laptop; peak memory below 2 GB.

each evaluation is a finite-simulation approximation whose characteristics enter derivative-based estimation (Section 2).

This paper concerns the factor-structured case

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i} \varepsilon_i, \quad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0, 1) \text{ i.i.d.}, \quad D_i > 0, \quad (1)$$

i.e. $\Sigma = VV^\top + \text{diag}(D)$ — the form in which applied covariance structure often arrives. Factor dimension reduction for probit is itself classical: one-factor quadrature goes back to Butler and Moffitt [4], and factor-analytic probit to Elrod and Keane [5]. What appears not to have been assembled is the *all-alternatives* computation: conditionally on the factors, a lattice survival-product identity [9] computes every alternative’s one-dimensional integral in a single $O(NL)$ pass — the field survival product is built once and each alternative’s leave-one-out field is recovered by division. A k -dimensional quadrature around that inner loop gives the whole share vector deterministically. The contribution here is that assembly, its inverse (share-to-utility) transform, the reuse of the conditional field for removal counterfactuals, and benchmark plus boundary evidence.

Proposition 1 (Conditional identity). *Under (1), with $m_i(f)$ the conditional location of alternative i ’s (sign-flipped, min-wins) performance and $S_j(x | f)$ its conditional survival function,*

$$p_i = \mathbb{E}_f \left[\int \frac{1}{\sqrt{D_i}} \phi\left(\frac{x - m_i(f)}{\sqrt{D_i}}\right) \prod_{j \neq i} S_j(x | f) dx \right], \quad \sum_i p_i = 1, \quad (2)$$

and the product over $j \neq i$, for all i simultaneously, is obtained from $\log S_{\text{field}} = \sum_j \log S_j$ by subtraction.

Proposition 2 (Complexity). *With Q factor nodes and L lattice points, the forward share vector costs $O(QNL)$; the full single-removal ensemble $\{P(j \text{ chosen} | i \text{ removed})\}_{i,j}$ costs $O(QN^2L)$ arithmetic and $O(N^2)$ storage, reusing one conditional field construction.*

Proposition 3 (Identification and differentiability). *$p(\mu + c\mathbf{1}) = p(\mu)$, so $J = \partial p / \partial \mu$ has $\mathbf{1}$ in its null space and μ is identified only up to location. With B an $N \times (N - 1)$ basis of $\mathbf{1}^\perp$ and $\mu = B\eta$, share inversion and implicit differentiation are taken in reduced coordinates: $d\eta/d\theta = -(B^\top JB)^{-1} B^\top \partial p / \partial \theta$.*

Only the factors need be Gaussian: (2) requires conditional independence and an integrable factor law, so Student, mixture, or empirical factor distributions are admissible with appropriate nodes, and the base densities may be arbitrary and alternative-specific.

Numerical method, stated completely. The benchmarks use $L = 1501$ lattice points on the adaptive interval $[\min_{i,f} m_i(f) - 8 \max_i \sqrt{D_i}, \max_{i,f} m_i(f) + 8 \max_i \sqrt{D_i}]$; product Gauss–Hermite nodes of order 15/11/9 for $k = 2/3/4$ with weights pruned below 10^{-7} of the maximum ($Q \approx 145$ at $k = 2$); scrambled-Sobol nodes (2^{13} points, fixed seed) for $k > 4$ — fixed-seed randomized QMC, disclosed as such; survival floors at 10^{-300} ; and normalization of the quadrature output by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2); statements about derivatives below refer to derivatives of this approximation, whose fidelity to exact MNP derivatives is governed by L and Q .

Inversion. Given target shares $\hat{p}$ and (V, D) , the inverse transform finds mean-zero μ with $p(\mu) = \hat{p}$ by a damped Jacobi-style diagonal-Newton iteration: all coordinates updated simultaneously using own-location slopes of the unnormalized lattice map (available from the same pass), the field refrozen each iteration, a warm start from the independent-field inverse using total variances, and a tail-aware tolerance (alternatives with target shares below $10^{-4}/N$ are matched best-effort; their locations are weakly identified — itself a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear). Convergence in a handful of iterations is an empirical observation on the tested problems, not a theorem. The design follows the original transform’s inversion [9] and the calibration practice of [14].

2 Benchmark against GHK

Anchors first. On the binary case, where MNP has a closed form, the lattice transform agrees to 3×10^{-16} and our GHK implementation — validated before being compared against — to 2×10^{-14} at $R = 2 \times 10^5$ (an easy anchor for both methods, useful for code validation rather than as evidence about the hard regime). At $N = 5$ both methods sit within the noise of a 10^7 -draw Monte Carlo reference, and the shipped `thurstone.FactorRace` agrees with the benchmark kernel to 1.1×10^{-5} .

Error metric: throughout, $\max_i |\hat{p}_i - p_i|$ against a simulation reference. A maximum over N coordinates tends to grow with N at fixed per-coordinate accuracy, and the reference’s own maximum-coordinate noise behaves likewise; mean-absolute and total-variation summaries accompany the repository output.

N	lattice transform	GHK ($R = 1000$)
5	21 ms, err 4.1×10^{-4}	1 ms, err 3.1×10^{-3}
20	95 ms, err 3.8×10^{-4}	12 ms, err 4.4×10^{-3}
50	257 ms, err 2.1×10^{-4}	77 ms, err 7.7×10^{-3}
200	930 ms, err 2.8×10^{-4}	1333 ms, err 6.7×10^{-3}
1000	4.5 s, err $8.0 \times 10^{-4} \dagger$	—
5000	22.6 s, err $9.0 \times 10^{-4} \dagger$	—

Table 1: Full share vector, $k = 2$ factors. Reference: 2×10^6 MC draws through $N = 200$; 5×10^5 draws at $N = 1000$ and 5000 ($\dagger$: there the reference’s own maximum-coordinate noise is of comparable order, so these entries bound the lattice error only to that order; replicated references are queued). Lattice error stayed below 10^{-3} throughout the tested sequence; GHK cost crossed over by $N \approx 200$.

Matched-accuracy extrapolation (not a measurement). Fitting a power law to the reference implementation’s timings over $N \in [20, 200]$ ($\alpha \approx 2.0$; a lower bound, the asymptotic per-share cost being cubic) and combining with $R^{-1/2}$ error scaling suggests that matching the lattice’s 9×10^{-4} at $N = 5000$ would take this implementation on the order of twelve hours, against 22 seconds. The baseline is a straightforward single-threaded Python GHK: per-alternative Cholesky, plain pseudorandom uniforms, one realization per problem, shares normalized by their sum. It establishes the scaling of the per-alternative simulation approach, not a field-wide impossibility; stronger baselines — direct utility simulation (whole share vector

per draw), QMC-GHK, seed-replicated uncertainty bands, and modern Gaussian-probability methods such as minimax tilting [8] — are queued in the repository.

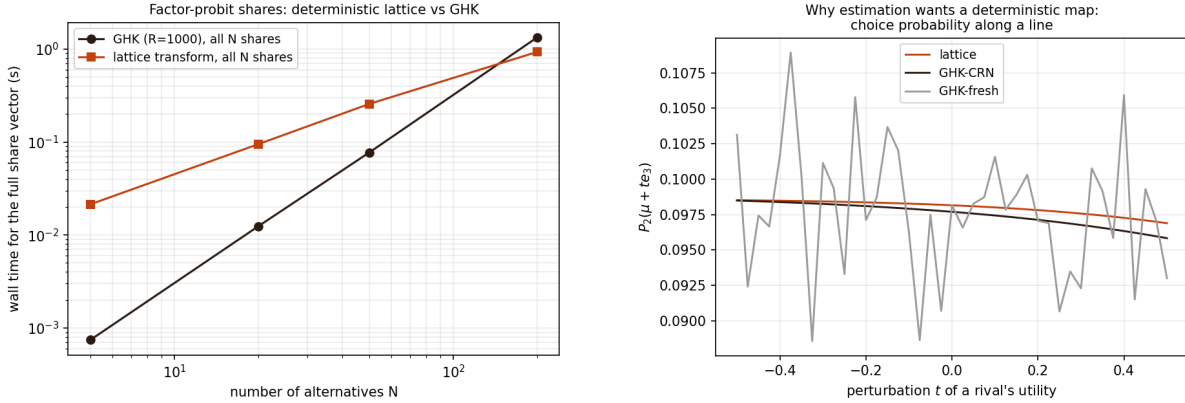


Figure 1: Left: wall time for the full share vector (log–log). Right: a choice probability along a line in utility space. Fresh-draw GHK shows run-to-run simulation noise (curvature statistic 39.3 vs. 0.01); common-random-number GHK is a smooth deterministic approximation conditional on its draw set, its finite- R error visible as the offset of its curve; the lattice curve is the smooth deterministic approximation of this paper. All three are approximations; a quantitative derivative-accuracy comparison against an independent high-accuracy reference at matched budgets is queued.

Derivatives, stated correctly. For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of μ whose derivatives are evaluated without resampling; analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws [2, 3]. The practical distinction is the character and controllability of the approximation error — quadrature resolution versus draw count — not the possibility of differentiation.

Share inversion. At $N = 1000$, $k = 2$, with targets from 5×10^6 MC draws floored at 10^{-7} and renormalized: the inversion completes in 59 seconds. The iteration solves its own equations to 6.3×10^{-9} — a statement about the solver, not statistical accuracy, since the target carries sampling noise $\sim 2 \times 10^{-4}$ — and recovers the utilities of identified alternatives (target share above 3×10^{-4}) to maximum error 0.014. Replication over independent target simulations and recovery-versus-share curves accompany the repository output. We are not aware of probit share inversion at this scale in current simulation-based practice.

Removal ensemble. All 200 single-removal share vectors from one conditional field construction in $O(QN^2L)$: $3.7\times$ faster than an extrapolation from direct recomputations, with ten evenly spaced removals checked directly (maximum discrepancy 5.6×10^{-17}).

3 Boundary experiments

General covariance through the factor approximation. The method requires $\Sigma \approx VV^\top + D$; GHK does not. We generated one random correlation matrix at each spectral decay $\lambda_m \propto m^{-\gamma}$, $\gamma \in \{0.5, 1.5, 3\}$, at $N = 50$ (single realizations; replication over matrices and utility vectors is queued), fit factors by iterated principal factor analysis, and measured share error against an 8×10^6 -draw reference, with GHK at the exact Σ for comparison. The plotted quantity is the *rank- k approximation error of the stated fitting and integration procedure* — covariance-fit error plus factor-integration and lattice error — not a minimum over factor models. Findings as they fell: error decays fastest for near-low-rank truth ($\gamma = 3$: 5.6×10^{-2} at $k = 1$ to 1.2×10^{-3} at $k = 8$) and slowest at intermediate decay ($\gamma = 1.5$: 2.4×10^{-2} to 2.6×10^{-3}); at every tested decay the $k = 8$ result (fixed-seed RQMC nodes) matched or beat GHK at $R = 10^4$ (3.2 – 4.1×10^{-3}). We expected a clear GHK-wins regime at this size and did not find one; where accuracy demands exceed the achievable rank- k error — intermediate spectra with k capped, or short-range correlation at large N — GHK’s arbitrary- Σ generality is the right tool. Further practical boundaries (small $D_{\min}/D_{\max}$, wide utility ranges, near-perfect correlation, extreme shares) stress the uniform lattice and are queued as resolution sweeps in L and Q .

Substitution fidelity, with noise families standardized. Why want probit-style flexibility at scale? Ground truth misspecified for every candidate: $t(5)$ -distributed factors and skew-normal idiosyncratic noise, *all noise families standardized to mean zero and unit variance*. (An earlier version of this experiment confounded noise family with noise variance; the standardized rerun is reported here, and the ordering survived.) Candidates — plain logit, factor mixed logit (unit-variance Gumbel base), factor probit — are calibrated to identical full-menu shares with the true loadings supplied, and scored on deletion counterfactuals against fresh simulation truths. Blocks whose removed share sits at the Monte Carlo noise floor are uninformative (all models tie); on informative blocks, the misallocated fraction of redistributed mass is:

	deleted mass > 10%	deleted mass 2–10%
plain logit (IIA)	15.9%	25.7%
factor mixed logit	7.7%	17.1%
factor probit	2.8%	7.8%

In this calibration, factor structure supplies the larger share of the improvement over IIA and the idiosyncratic family a further improvement — with the caveats that the truth’s skew-normal noise is Gaussian-adjacent (favoring probit) and that this is a single design and seed; replication across skewness and tail designs is queued before any general claim.

4 Related work

GHK and simulated likelihood are standard [1, 2, 3]; analytic-approximation lines include MACML [6]; modern Gaussian-probability methods include minimax tilting [8]. Factor dimension reduction for probit is classical: Butler–Moffitt one-factor quadrature [4], Elrod–Keane factor-analytic probit [5], and the probit probability approximations surveyed in [7]. The lattice survival-product identity and its inverse transform are from [9]. The novelty claimed here

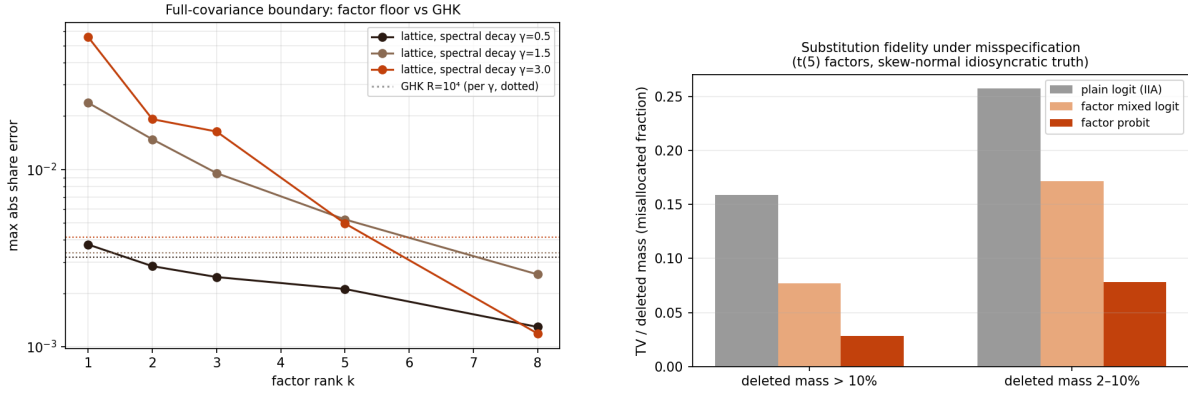


Figure 2: Left: rank- k approximation error under spectral decay γ (dotted: GHK at $R = 10^4$, per γ). Right: substitution fidelity with standardized noise families, stratified by deleted share mass.

is the all-alternatives assembly with inversion and removal-ensemble reuse — not dimensional reduction, which is old. Share inversion and its acceleration are central to demand estimation [10, 11]; automatic differentiation of logit-based BLP is recent [12]; a 2026 preprint approximates correlated-choice probabilities with equivariant neural networks [13], evidence of current demand for this computation. We are unaware of an automatically differentiable, large-choice-set factor-probit demand implementation built on deterministic share inversion; that precise object is what Section 5 proposes.

5 Discussion

Estimation is the natural next step: with the reduced coordinates of Proposition 3, gradients of an outer likelihood or GMM objective pass through the share-inversion fixed point by implicit differentiation, a candidate route to differentiable probit demand estimation at large N . Estimating (V, D) rather than supplying them raises the usual identification cautions: utility scale normalization, factor rotation, choice-invariant common loading components, and the fact that only difference covariances are identified. Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities; a *common-scale* Gumbel base with zero loadings reproduces the Luce model exactly (alternative-specific scales break Luce, as they should), yielding correlated-softmax generalizations. Finally, the inversion inherits the original transform’s interpolation structure (per-node location-to-probability curves are cross-correlations, evaluable at all offsets simultaneously), which we have not yet exploited.

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